

3 Meromorphic Functions and the Logarithm

One knows that the differential calculus, which has contributed so much to the progress of analysis, is founded on the consideration of differential coefficients, that is derivatives of functions. When one attributes an infinitesimal increase ϵ to the variable x , the function $f(x)$ of this variable undergoes in general an infinitesimal increase of which the first term is proportional to ϵ , and the finite coefficient of ϵ of this increase is what is called its differential coefficient... If considering the values of x where $f(x)$ becomes infinite, we add to one of these values designated by x_1 , the infinitesimal ϵ , and then develop $f(x_1 + \epsilon)$ in increasing power of the same quantity, the first terms of this development contain negative powers of ϵ ; one of these will be the product of $1/\epsilon$ with a finite coefficient, which we will call the residue of the function $f(x)$, relative to the particular value x_1 of the variable x . Residues of this kind present themselves naturally in several branches of algebraic and infinitesimal analysis. Their consideration furnish methods that can be simply used, that apply to a large number of diverse questions, and that give new formulae that would seem to be of interest to mathematicians...

A. L. Cauchy, 1826

There is a general principle in the theory, already implicit in Riemann's work, which states that analytic functions are in an essential way characterized by their singularities. That is to say, globally analytic functions are “effectively” determined by their zeros, and meromorphic functions by their zeros and poles. While these assertions cannot be formulated as precise general theorems, there are nevertheless significant instances where this principle applies.

We begin this chapter by considering singularities, in particular the different kind of point singularities (“isolated” singularities) that a holomorphic function can have. In order of increasing severity, these are:

- removable singularities

- poles
- essential singularities.

The first type is harmless since a function can actually be extended to be holomorphic at its removable singularities (hence the name). Near the third type, the function oscillates and may grow faster than any power, and a complete understanding of its behavior is not easy. For the second type the analysis is more straight forward and is connected with the calculus of residues, which arises as follows.

Recall that by Cauchy's theorem a holomorphic function f in an open set which contains a closed curve γ and its interior satisfies

$$\int_{\gamma} f(z) dz = 0.$$

The question that occurs is: what happens if f has a pole in the interior of the curve? To try to answer this question consider the example $f(z) = 1/z$, and recall that if C is a (positively oriented) circle centered at 0, then

$$\int_C \frac{dz}{z} = 2\pi i.$$

This turns out to be the key ingredient in the calculus of residues.

A new aspect appears when we consider indefinite integrals of holomorphic functions that have singularities. As the basic example $f(z) = 1/z$ shows, the resulting “function” (in this case the logarithm) may not be single-valued, and understanding this phenomenon is of importance for a number of subjects. Exploiting this multi-valuedness leads in effect to the “argument principle.” We can use this principle to count the number of zeros of a holomorphic function inside a suitable curve. As a simple consequence of this result, we obtain a significant geometric property of holomorphic functions: they are open mappings. From this, the maximum principle, another important feature of holomorphic functions, is an easy step.

In order to turn to the logarithm itself, and come to grips with the precise nature of its multi-valuedness, we introduce the notions of homotopy of curves and simply connected domains. It is on the latter type of open sets that single-valued branches of the logarithm can be defined.

1 Zeros and poles

By definition, a **point singularity** of a function f is a complex number z_0 such that f is defined in a neighborhood of z_0 but not at the point

z_0 itself. We shall also call such points **isolated singularities**. For example, if the function f is defined only on the punctured plane by $f(z) = z$, then the origin is a point singularity. Of course, in that case, the function f can actually be defined at 0 by setting $f(0) = 0$, so that the resulting extension is continuous and in fact entire. (Such points are then called removable singularities.) More interesting is the case of the function $g(z) = 1/z$ defined in the punctured plane. It is clear now that g cannot be defined as a continuous function, much less as a holomorphic function, at the point 0. In fact, $g(z)$ grows to infinity as z approaches 0, and we shall say that the origin is a pole singularity. Finally, the case of the function $h(z) = e^{1/z}$ on the punctured plane shows that removable singularities and poles do not tell the whole story. Indeed, the function $h(z)$ grows indefinitely as z approaches 0 on the positive real line, while h approaches 0 as z goes to 0 on the negative real axis. Finally h oscillates rapidly, yet remains bounded, as z approaches the origin on the imaginary axis.

Since singularities often appear because the denominator of a fraction vanishes, we begin with a local study of the zeros of a holomorphic function.

A complex number z_0 is a **zero** for the holomorphic function f if $f(z_0) = 0$. In particular, analytic continuation shows that the zeros of a non-trivial holomorphic function are isolated. In other words, if f is holomorphic in Ω and $f(z_0) = 0$ for some $z_0 \in \Omega$, then there exists an open neighborhood U of z_0 such that $f(z) \neq 0$ for all $z \in U - \{z_0\}$, unless f is identically zero. We start with a local description of a holomorphic function near a zero.

Theorem 1.1 *Suppose that f is holomorphic in a connected open set Ω , has a zero at a point $z_0 \in \Omega$, and does not vanish identically in Ω . Then there exists a neighborhood $U \subset \Omega$ of z_0 , a non-vanishing holomorphic function g on U , and a unique positive integer n such that*

$$f(z) = (z - z_0)^n g(z) \quad \text{for all } z \in U.$$

Proof. Since Ω is connected and f is not identically zero, we conclude that f is not identically zero in a neighborhood of z_0 . In a small disc centered at z_0 the function f has a power series expansion

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k.$$

Since f is not identically zero near z_0 , there exists a smallest integer n

such that $a_n \neq 0$. Then, we can write

$$f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z),$$

where g is defined by the series in brackets, and hence is holomorphic, and is nowhere vanishing for all z close to z_0 (since $a_n \neq 0$). To prove the uniqueness of the integer n , suppose that we can also write

$$f(z) = (z - z_0)^m g(z) = (z - z_0)^m h(z)$$

where $h(z_0) \neq 0$. If $m > n$, then we may divide by $(z - z_0)^n$ to see that

$$g(z) = (z - z_0)^{m-n} h(z)$$

and letting $z \rightarrow z_0$ yields $g(z_0) = 0$, a contradiction. If $m < n$ a similar argument gives $h(z_0) = 0$, which is also a contradiction. We conclude that $m = n$, thus $h = g$, and the theorem is proved.

In the case of the above theorem, we say that f has a **zero of order** n (or **multiplicity** n) at z_0 . If a zero is of order 1, we say that it is **simple**. We observe that, quantitatively, the order describes the rate at which the function vanishes.

The importance of the previous theorem comes from the fact that we can now describe precisely the type of singularity possessed by the function $1/f$ at z_0 .

For this purpose, it is now convenient to define a **deleted neighborhood** of z_0 to be an open disc centered at z_0 , minus the point z_0 , that is, the set

$$\{z : 0 < |z - z_0| < r\}$$

for some $r > 0$. Then, we say that a function f defined in a deleted neighborhood of z_0 has a **pole** at z_0 , if the function $1/f$, defined to be zero at z_0 , is holomorphic in a full neighborhood of z_0 .

Theorem 1.2 *If f has a pole at $z_0 \in \Omega$, then in a neighborhood of that point there exist a non-vanishing holomorphic function h and a unique positive integer n such that*

$$f(z) = (z - z_0)^{-n} h(z).$$

Proof. By the previous theorem we have $1/f(z) = (z - z_0)^n g(z)$, where g is holomorphic and non-vanishing in a neighborhood of z_0 , so the result follows with $h(z) = 1/g(z)$.

The integer n is called the **order** (or **multiplicity**) of the pole, and describes the rate at which the function grows near z_0 . If the pole is of order 1, we say that it is **simple**.

The next theorem should be reminiscent of power series expansion, except that now we allow terms of negative order, to account for the presence of a pole.

Theorem 1.3 *If f has a pole of order n at z_0 , then*

$$(1) \quad f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{(z - z_0)} + G(z),$$

where G is a holomorphic function in a neighborhood of z_0 .

Proof. The proof follows from the multiplicative statement in the previous theorem. Indeed, the function h has a power series expansion

$$h(z) = A_0 + A_1(z - z_0) + \cdots$$

so that

$$\begin{aligned} f(z) &= (z - z_0)^{-n}(A_0 + A_1(z - z_0) + \cdots) \\ &= \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{(z - z_0)} + G(z). \end{aligned}$$

The sum

$$\frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{(z - z_0)}$$

is called the **principal part** of f at the pole z_0 , and the coefficient a_{-1} is the **residue** of f at that pole. We write $\text{res}_{z_0} f = a_{-1}$. The importance of the residue comes from the fact that all the other terms in the principal part, that is, those of order strictly greater than 1, have primitives in a deleted neighborhood of z_0 . Therefore, if $P(z)$ denotes the principal part above and C is any circle centered at z_0 , we get

$$\frac{1}{2\pi i} \int_C P(z) dz = a_{-1}.$$

We shall return to this important point in the section on the residue formula.

As we shall see, in many cases, the evaluation of integrals reduces to the calculation of residues. In the case when f has a simple pole at z_0 , it is clear that

$$\text{res}_{z_0} f = \lim_{z \rightarrow z_0} (z - z_0)f(z).$$

If the pole is of higher order, a similar formula holds, one that involves differentiation as well as taking a limit.

Theorem 1.4 *If f has a pole of order n at z_0 , then*

$$\operatorname{res}_{z_0} f = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z - z_0)^n f(z).$$

The theorem is an immediate consequence of formula (1), which implies

$$(z - z_0)^n f(z) = a_{-n} + a_{-n+1}(z - z_0) + \cdots + a_{-1}(z - z_0)^{n-1} + \\ + G(z)(z - z_0)^n.$$

2 The residue formula

We now discuss the celebrated residue formula. Our approach follows the discussion of Cauchy's theorem in the last chapter: we first consider the case of the circle and its interior the disc, and then explain generalizations to toy contours and their interiors.

Theorem 2.1 *Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C . Then*

$$\int_C f(z) dz = 2\pi i \operatorname{res}_{z_0} f.$$

Proof. Once again, we may choose a keyhole contour that avoids the pole, and let the width of the corridor go to zero to see that

$$\int_C f(z) dz = \int_{C_\epsilon} f(z) dz$$

where C_ϵ is the small circle centered at the pole z_0 and of radius ϵ .

Now we observe that

$$\frac{1}{2\pi i} \int_{C_\epsilon} \frac{a_{-1}}{z - z_0} dz = a_{-1}$$

is an immediate consequence of the Cauchy integral formula (Theorem 4.1 of the previous chapter), applied to the constant function $f = a_{-1}$. Similarly,

$$\frac{1}{2\pi i} \int_{C_\epsilon} \frac{a_{-k}}{(z - z_0)^k} dz = 0$$